

Reading Unseen Scripts: Predictable Zero-Real-Image Transfer for Brahmic Scene Text Recognition

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Abstract

001 Scene-text recognition (STR) is effectively solved for
002 Latin script yet fails for most of the world’s writing
003 systems, and collecting labeled real images for every
004 script is not a scalable answer. We ask a stronger
005 question: can a recognizer read a script for which no
006 real labeled image exists at all — and can its accuracy
007 be predicted before training? We construct a shared
008 grapheme pivot space spanning nine Brahmic scripts
009 and fine-tune a compact open vision-language model
010 in it under a leave-one-script-out protocol: the held-out
011 script is seen only as rendered synthetic images, never
012 as a real photograph. On real scene-text benchmarks
013 the held-out scripts reach 9.16–30.09% word recognition
014 (character accuracy up to 57.77%), against source-
015 only baselines of 0.0–5.46%. Replacing the grapheme
016 pivot with the stock BPE tokenizer degrades transfer
017 on every script (up to 3.9×) at comparable character
018 accuracy, showing that the pivot, not fine-tuning alone,
019 converts character evidence into word reads. tokenizer
020 fertility predicts which scripts transfer — was falsified
021 (Spearman -0.80) and is reported as such; the surviv-
022 ing two-factor account (synthetic-exposure budget and
023 pivot-space coverage) was itself tested prospectively,
024 with point predictions and error bands committed to
025 a version-controlled archive before each remaining run
026 and scored as hits and misses. We release the pivot con-
027 struction, rendering pipeline, and the zero-real-image
028 evaluation protocol.

029 1. Introduction

030 Modern scene-text recognizers exceed 90% word accu-
031 racy on Latin benchmarks, yet most of the world’s
032 writing systems remain unreadable to them. Recent
033 audits are stark: across more than a hundred Unicode
034 scripts, even frontier vision-language models (VLMs)
035 read fewer than thirty, and on unfamiliar scripts they

emit garbled or substituted output [14]. For the Brah-
mic family — nine major scripts used by over a billion
people — real-image training data exists only because
of recent, labor-intensive benchmark efforts [8, 19], and
dozens of related scripts (Sinhala, Khmer, Myanmar,
Tibetan, ...) have little or none. Collecting and anno-
tating real photographs script-by-script is not a scal-
able path to coverage.

This paper asks a stronger question than “how well
can script X be read with its training set”: can a rec-
ognizer read a script for which no real labeled image
exists at all — and can we predict, before training, how
well? We answer both, for scene text, at the granular-
ity of an entire writing system.

Our approach rests on a structural observation
about Brahmic scripts: they descend from a common
ancestor and share an organizational scheme — conso-
nant bases, vowel signs, and conjunct formation via a
virama — even though their glyphs and Unicode blocks
are disjoint. We exploit this by constructing a shared
grapheme pivot space: every script’s text is decom-
posed into grapheme clusters and mapped by structural
correspondence into a common output vocabulary, so
that nine visually unrelated scripts write into one label
space. A compact open VLM (Florence-2, 0.23B pa-
rameters [24]) is fine-tuned in this space on real images
of source scripts plus rendered synthetic images of the
target script — never a real target photograph.

We evaluate with a leave-one-script-out (LOSO)
protocol over all nine scripts of a public real-image
benchmark [8]. For each held-out script we train
two models: a source-only model (Rung A), and a
source+synthetic-target model (Rung B); both are
evaluated on the held-out script’s real test images.
Transfer is substantial and universal: every script im-
proves from a 0.0–5.46% baseline to 9.16–30.09% word
recognition rate (WRR), with character accuracy up
to 57.77% — using zero real target images. An abla-
tion that swaps the grapheme pivot for the stock BPE

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075 tokenizer, all else identical, degrades transfer on every
076 script observed (up to $3.9\times$); strikingly, on Telugu the
077 BPE model matches the pivot model's character accu-
078 racy (55.1% vs. 54.43%) while halving word accuracy
079 — the pivot is what converts character evidence into
080 whole-word reads.

081 The second contribution is epistemic, and to our
082 knowledge unprecedented in text recognition: the cam-
083 paign was run under a frozen, version-controlled pre-
084 registration with an append-only amendment log, and
085 its predictions were filed prospectively — committed
086 and pushed before the runs they predict. This cuts
087 both ways, and we report both edges. Our prereg-
088 istered primary hypothesis — that tokenizer fertility,
089 the property that governs the supervised grapheme-
090 vs-BPE trade-off (Sec. 3), also predicts which scripts
091 transfer — was falsified (Spearman -0.80 across nine
092 scripts). The account that survived is a two-factor
093 one: synthetic-exposure budget dominates, and within
094 a fixed budget, pivot-space coverage orders transfer.
095 Its out-of-sample record is auditable: a point forecast of
096 ~ 15 WRR for Kannada (realized 15.42); a final-script
097 forecast for Gurmukhi of 16.2 WRR with $\pm 2\sigma$ band
098 $[8.2, 24.1]$, filed and pushed 3.7 hours before training
099 began (realized 22.16: outer band hit, point under-
100 called — reported as the miss it is).

101 Our contributions:

- 102 1. Zero-real-image cross-script transfer. The first
103 demonstration, to our knowledge, that real scene
104 text of an entirely unseen script — disjoint Uni-
105 code block, disjoint glyph inventory — can be read
106 by transferring through a designed structural pivot
107 space, evaluated LOSO across nine Brahmic scripts
108 (9.16–30.09% WRR from $\leq 5.46\%$ baselines).
- 109 2. The pivot is necessary, not incidental. A prereg-
110 istered stock-BPE ablation under identical data, seed,
111 and recipe degrades WRR on every script (up to
112 $3.9\times$), isolating output-space structure — not syn-
113 thetic fine-tuning per se — as the transfer mecha-
114 nism.
- 115 3. Predictability, tested prospectively. A two-factor
116 model (synthetic budget + pivot coverage) with
117 committed-before-run forecasts, scored hits and
118 misses, a falsified preregistered primary reported
119 in full, and a dose-response synthetic-scaling study
120 (Sec. 4).
- 121 4. Artifacts. The pivot construction, rendering and
122 verification pipeline, evaluation protocol, and the
123 preregistration/prediction chain itself, released as a
124 reusable template for prospective empirical ML.

125 We do not claim per-script state of the art: super-
126 vised recognizers trained on real labeled target images
127 reach $\sim 73\%$ on this benchmark [8] and sit on a differ-

ent axis — they presuppose the data whose absence
defines our problem. The relevant comparisons, which
we report, are source-only baselines, the BPE ablation,
raw and frontier VLMs on the same test sets, and our
own prospective forecasts.

2. Related Work

Multilingual STR and Indic benchmarks. STR is ma-
ture for Latin script [3], but coverage collapses else-
where. IndicSTR12 [19] and the Bharat Scene Text
Dataset [8] established real-image benchmarks for In-
dian languages; on the latter, PARSeq reaches $\sim 47\%$
average WRR trained on synthetic data only and
 $\sim 73\%$ after fine-tuning on real labeled images —
numbers that presuppose labeled target data. Glo-
tOCR [14] quantifies the wider problem: frontier VLMs
read fewer than thirty of 100+ scripts. Synthetic-data
engines for STR remain Latin-centric [25], and fine-
tuning VLMs on synthetic renderings has been shown
for a single low-resource script with real-data valida-
tion [6]. Our work targets the regime where no real
labeled target image exists, and asks not only whether
such scripts can be read but how well, predicted in
advance.

Tokenization granularity for complex scripts. MGP-
STR [7] fuses character, BPE, and WordPiece streams
within one Latin recognizer; grapheme-level tokeniza-
tion beats byte-BPE for Bengali handwriting [11]. In
the LLM literature, fertility's value as a tokenizer met-
ric is contested [20]. Our preliminary study (Sec. 3)
makes the supervised grapheme-vs-BPE choice pre-
dictable across nine scripts (polarity 8/9, leave-one-
script-out 90.9%), and this paper then tests — and
refutes — the natural conjecture that the same prop-
erty predicts transfer.

Unseen units vs. unseen scripts. A substantial line
reads unseen characters within a known script by de-
composition into shared sub-units: stroke- and radical-
based zero-shot Chinese recognition [4, 26, 27], and
open-set recognition of novel characters [16–18]. Zero-
shot detection of unseen scripts localizes but does not
read [15]. Byte-level pivots let ASR cover languages
with unseen scripts in speech [5], and task-arithmetic
analogies achieve zero-real-data synthetic-to-real HTR
— for five Latin-script languages, with unseen writing
systems explicitly left open [10]. To our knowledge, no
prior system reads real scene text of an entirely unseen
script — different Unicode block, different glyphs —
from zero real target images, let alone with accuracy
forecast beforehand.

177	What predicts cross-lingual transfer. In NLP, trans-	224
178	fer tracks lexical and subword overlap rather than lan-	225
179	guage relatedness [9, 22]. For STR specifically, Ref. [1]	226
180	concludes that source dataset size and visual similar-	227
181	ity, not linguistic typology, drive cross-lingual trans-	228
182	fer. Our findings sharpen rather than contradict this,	229
183	in a regime that work did not study (zero real tar-	230
184	get images): (i) a typology-adjacent structural prop-	231
185	erty — fertility — indeed fails as a transfer predic-	232
186	tor (-0.80), as a data-centric view expects; (ii) syn-	233
187	thetic target exposure dominates — a target-side data-	234
188	size effect; (iii) within a fixed budget, coverage of the	235
189	shared grapheme space orders transfer ($+0.61$ among	236
190	equal-budget scripts) — an OCR instantiation of the	237
191	subword-overlap principle in a designed output space;	238
192	and (iv) a frozen-encoder visual-similarity control pre-	239
193	dicts nothing in our data (-0.14). Because LOSO bal-	240
194	ances source data by construction, the confound iden-	
195	tified by [1] is controlled by design.	
196	Prediction and preregistration in empirical ML. Scal-	
197	ing laws make supervised STR predictable in model	
198	and data size [23]; preregistration and negative-result	
199	reporting are the reproducibility literature’s recom-	
200	ended remedies for HARKing and cherry-picking [12,	
201	13], yet full-study adoption is rare. We run the en-	
202	tire campaign under a frozen preregistration with an	
203	append-only amendment log and commit quantitative	
204	forecasts before the runs they predict; we offer the pro-	
205	cedure itself as a contribution.	
206	3. Method	
207	3.1. A shared grapheme pivot space for Brahmic	
208	scripts	
209	Brahmic scripts encode the same phonological skeleton	
210	— consonant bases, dependent vowel signs, and con-	
211	junct formation through a virama — and Unicode pre-	
212	serves this: corresponding letters sit at the same offset	
213	within each script’s block. We exploit both layers.	
214	Structural pivot. A mapping π sends any Brahmic	
215	codepoint to a pivot codepoint by its within-block off-	
216	set ($\pi(c) = 0x0900 + (c - \text{base}(c))$); shared punctua-	
217	tion that is not offset-safe is excluded by construction.	
218	Because π is a per-script bijective shift, it is exactly	
219	invertible: a hypothesis decoded in pivot space is read	
220	back into the target script deterministically. Text from	
221	nine scripts thereby writes into one label space in which	
222	structurally corresponding syllables become identical	
223	strings.	
	Grapheme-cluster vocabulary. Within pivot space	224
	we segment text into grapheme clusters (consonant-	225
	virama conjuncts kept whole, dependent signs at-	226
	tached) and inject the resulting cluster inventory	227
	(1,624 types for the source pool) as dedicated tokens	228
	into the recognizer’s tokenizer. The alternative — let-	229
	ting the stock byte-BPE tokenizer segment pivot-space	230
	text — is the ablation of Sec. 4: it sees the same	231
	pivot strings but fragments clusters into sub-syllabic	232
	pieces. Fertility (mean clusters per word) governs when	233
	grapheme tokenization beats BPE in supervised train-	234
	ing: in a 12-run balanced-budget study across the same	235
	nine scripts, fertility predicts the winning branch with	236
	8/9 polarity and 90.9% leave-one-script-out direction	237
	accuracy ($R^2=0.678$). That result motivates both the	238
	pivot design and the (ultimately falsified) primary hy-	239
	pothesis of Sec. 5.	240
	3.2. Zero-real-image protocol	241
	For each held-out script S of the nine in the bench-	242
	mark [8]:	243
	Sources. All benchmark languages whose script is	244
	not S (languages sharing S , e.g. Assamese for Bengali	245
	script, are excluded — S is genuinely unseen). Each	246
	source language contributes exactly 700 train / 90 val	247
	real images (seed 42), giving balanced source exposure	248
	by construction; source-side data size is therefore con-	249
	trolled, unlike prior cross-lingual STR studies [1].	250
	Synthetic target. 3,240 images per target language:	251
	benchmark-vocabulary words rendered in fontconfig-	252
	verified fonts (multiple faces per script, libraqm shap-	253
	ing), each render checked non-blank; the rendered word	254
	enters training as its pivot transcription. No real pho-	255
	tograph of S is ever seen. Scripts written by two source	256
	languages receive two languages’ worth of renderings	257
	($2\times$ budget) — a covariate disclosed at preregistration	258
	time that becomes central in Sec. 5.	259
	Rungs. Rung A trains on sources only; Rung B adds	260
	the synthetic target set; Rung B _{bpe} repeats Rung B	261
	with the grapheme-cluster injection removed (stock to-	262
	kenizer), identical data, seed, and recipe. All rungs	263
	fine-tune Florence-2 (0.23B) [24] with LoRA ($r=16$,	264
	$\alpha=32$, dropout 0.05), lr 10^{-4} , batch size 4, 15 epochs,	265
	validation-selected checkpoint, beam-search decoding	266
	at test.	267
	Evaluation. The held-out script’s real test split, WRR	268
	= exact match after NFC normalization, zero-width-	269
	character removal, and whitespace collapse — the	270
	benchmark’s own protocol [8, 19] — plus character ac-	271
	curacy ($1 - \text{CER}$).	272

[TODO: Fig. 1: pivot-space diagram — nine scripts, offset maps into shared space, grapheme-cluster vocabulary, LOSO rungs A/B/B_{bpe}]

Figure 1. The shared grapheme pivot space and the leave-one-script-out protocol. Each Brahmic script maps bijectively into a common output space by Unicode-offset correspondence; grapheme clusters in that space form the recognizer’s output vocabulary. The held-out script enters training only as verified synthetic renderings (Rung B) or not at all (Rung A); evaluation is always on its real scene images.

3.3. Preregistration and prospective predictions

The campaign ran under a frozen preregistration with an append-only amendment log; every amendment was filed before the results it concerns existed. Descriptors entering the transfer horse-race were computed result-blind: fertility and pivot coverage before any Rung-B result; a frozen-encoder visual-similarity control before the third. Quantitative forecasts for later runs were committed and pushed to a version-controlled archive before those runs trained (commit hashes in the supplementary material; archive anonymized for review), and are scored against a rule fixed at filing time — hits and misses alike (Sec. 5).

4. Experiments

Before any absolute number: supervised recognizers fine-tuned on real labeled target images reach $\sim 73\%$ WRR on this benchmark [8]. Those systems answer a different question — they presuppose the labeled data whose absence defines our setting. The comparisons that match our question are: the source-only baseline (Rung A), the raw pre-trained VLM (WRR 0.0), the stock-BPE ablation, frontier VLMs prompted zero-shot on the same test sets, and our own forecasts.

4.1. Main result: every unseen script becomes readable

Table 1 is the campaign’s core. Every one of the nine scripts moves from a near-floor source-only baseline (0.0–5.46% WRR; the raw pre-trained model reads none of them) to 9.16–30.09% WRR with character accuracy 34.57–57.77% — without a single real target image. Transfer does not collapse even on the hard-

Script	N	WRR (%)		Rung B	
		A	B	CharAcc	B _{bpe}
Tamil	513	0.0	9.16	39.0	2.34
Telugu	545	1.28	19.82	54.43	10.28
Kannada	720	2.78	15.42	46.37	13.33
Malayalam	547	0.18	9.69	34.57	4.2
Oriya	1044	0.77	25.38	56.37	14.94
Gujarati	1015	3.05	15.86	38.49	[TODO: run]
Bengali	2873	1.39	27.08	57.77	[TODO: run]
Devanagari	6042	5.46	30.09	57.16	[TODO: run]
Gurmukhi	2879	0.59	22.16	44.1	[TODO: run]

Table 1. Zero-real-image transfer across nine held-out Brahmic scripts (real test images). Rung A: source scripts only. Rung B: + synthetic target renderings, zero real target images. B_{bpe}: Rung B with the grapheme-cluster vocabulary removed (stock tokenizer), identical data and seed. [TODO: fill remaining B_{bpe} cells as rungs complete; add per-script 95% CIs (bootstrap over samples).]

est script (Malayalam, lowest pivot coverage at 79.2%: still 0.18 $\rightarrow$ 9.69). Rung-A baselines track incidental pivot-space coverage (highest for Devanagari, 5.46%), confirming the pivot shares structure before any target exposure.

4.2. The pivot is the mechanism: BPE ablation

Removing the grapheme-cluster vocabulary — same pivot strings, same images, same seed — degrades every script observed so far: Tamil 9.16 $\rightarrow$ 2.34 WRR (3.9 $\times$), Malayalam 9.69 $\rightarrow$ 4.2 (2.3 $\times$), Telugu 19.82 $\rightarrow$ 10.28 (1.9 $\times$), Oriya 25.38 $\rightarrow$ 14.94 (1.7 $\times$), Kannada 15.42 $\rightarrow$ 13.33 (1.2 $\times$) — the margin varies, the direction never does. The character-accuracy pattern is di-

[TODO: Fig. 2: nine-script grouped bars (A / B / B_{bpe}), CharAcc markers overlaid]

Figure 2. Transfer by script. [TODO: generate from result JSONs (make_wacv_figs.py).]

agnostic: on Telugu the BPE model matches the pivot model’s character accuracy (55.1% vs. 54.43%), and on Kannada slightly exceeds it (48.76% vs. 46.37%), yet word accuracy drops in every case. Synthetic fine-tuning alone teaches the model to see the characters; the cluster-level output space is what assembles them into correct whole words. [TODO: complete for the remaining four scripts (preregistered Amendment 3; runs in progress); report the paired per-script test and the final ratio range.]

4.3. Dose–response: synthetic budget

[TODO: Preregistered Amendment 4a: malayalam/kannada/telugu $\times$ {810, 1620, 3240, 6480, 12960} synthetic images, identical recipe; per-script $\text{WRR} = \alpha + \beta \log_2(\text{synth})$ and joint two-factor fit; test of the frozen $c \approx +11.48$ WRR-per-doubling prediction. Fig. 3 = dose–response curves. Runs queued after the ablation completes.]

4.4. Out-of-benchmark: Khmer

[TODO: Preregistered Amendment 4b: train on all nine benchmark scripts + synthetic Khmer (different Unicode block, coeng-based stacking, outside the benchmark and outside India); evaluate on KhmerST [21]. PROSPECTIVE_PREDICTION_KHMER filed from the frozen two-factor model before training. Timeboxed; paper stands without it if the segmenter/crop engineering misses the freeze date.]

4.5. Frontier-VLM reference

[TODO: Preregistered Amendment 4c: Qwen2.5-VL-7B [2] prompted zero-shot on the same nine real test sets, same normalization and metric. Declared caveat: unknown training data makes this an upper-bound reference for off-the-shelf capability, not a like-for-like comparison. Context: frontier models read fewer than thirty of 100+ scripts [14].]

5. What Predicts Transfer?

This section reports the campaign’s confirmatory analysis exactly as preregistered, including the falsification of our own primary hypothesis.

5.1. The preregistered primary fails

Fertility — the property that governs the supervised grapheme-vs-BPE trade-off (Sec. 3.1) — was preregistered as the primary transfer predictor (H3). It is refuted: Spearman $\rho = -0.80$ across the nine Rung-B results — directionally opposite to the hypothesis. The flagship case was called in advance by the declared competitor: our prospective prediction file ranked Malayalam best under fertility (P1) and worst under coverage (P2); it realized bottom-tier (9.69 WRR). We report this as a genuine falsification, consistent with fertility skepticism in the LLM-tokenizer literature [20] and with the data-centric view of STR transfer [1].

5.2. The two-factor account

Two result-blind quantities explain the campaign. (i) Synthetic budget dominates. The two scripts with the disclosed $2\times$ synthetic budget (Bengali, Devanagari) are the top two results (27.08, 30.09 WRR) — Bengali from the second-lowest coverage. (ii) Coverage orders transfer within a budget. Among the seven equal-budget scripts, pivot-space token coverage correlates at $\rho = +0.61$ (raw, all nine: $+0.37$). The joint fit $\text{WRR} = -35.79 + 0.583 \cdot \text{cov} + 11.48 \cdot \mathbb{I}[2\times]$ attains RMSE 4.18 WRR across all nine scripts, with both $2\times$ residuals near zero. A frozen-encoder visual-similarity control — rendered glyphs embedded by the recognizer’s own vision encoder — predicts nothing ($\rho = -0.14$), separating the structural coverage effect from the visual-similarity account of [1]. [TODO: report bootstrap CIs for both coefficients; cross-check -0.14 at 9 points.]

5.3. Prospective scorecard

Three forecasts were filed before their runs; all are scored (Fig. 3). Kannada: exploratory point ~ 15 WRR, filed with two scripts observed; realized 15.42. Devanagari: directional call (high, $\sim 25+$) committed before the result existed but pushed after — we therefore label it only weakly prospective and lean on it for nothing. Gurmukhi (final script): point 16.2 WRR, bands $\pm 1\sigma$ [12.2, 20.2] / $\pm 2\sigma$ [8.2, 24.1], committed and pushed 3.7 hours before training began; realized 22.16 — outside the $\pm 1\sigma$ band (under-called by 1.5σ), inside $\pm 2\sigma$. The declared rank bet between the two preregistered predictors resolved for coverage over fertility ($\rho +0.32$ vs. -0.71 on the seven forecast scripts), with neither ranking alone significant — which

[TODO: Fig. 4: prediction receipts — for each filed forecast, the committed point + $\pm 1\sigma / \pm 2\sigma$ bands vs. realized WRR, with commit-and-push timestamps relative to training start]

Figure 3. Every quantitative forecast filed during the campaign, scored against the rule fixed at filing time. Hits and misses are both shown.

is why the two-factor account, not a single-predictor law, is our claim. The model’s honest summary: transfer is forecastable to roughly ± 4.18 WRR (1σ) from two quantities computable before training — and the misses are part of the record.

6. Limitations

Absolute accuracy. 9.16–30.09% WRR reads many words but is far below the supervised ceiling ($\sim 73\%$ with real labeled data [8]); the synthetic $\rightarrow$ real domain gap is the dominant residual, and our training loss reaching ~ 0.06 on synthetic data while real-image WRR stays modest quantifies it. One architecture. All transfer results use Florence-2; the supervised granularity study spans 12 runs of a second architecture family, but architecture-generality of transfer is untested. One family. Nine scripts, one benchmark, one script family whose shared structure the pivot exploits by design; Khmer (Sec. 4.4) probes the boundary, and non-Brahmic generalization is open. Model imperfections. Gujarati underperforms its coverage (highest coverage, mid-table transfer), one Rung-A baseline breaches the predicted < 5 line by 0.46 (Devanagari, 5.46), and the final-script forecast under-called by 1.5σ ; all are reported, none are explained away. Compute. Single 24 GB GPU; budgets (700/source, 3,240 synthetic) were fixed by preregistration, not tuned.

7. Conclusion

We showed that real scene text of an entirely unseen Brahmic script can be read — at 9.16–30.09% word accuracy across all nine scripts of a public benchmark — by a small open VLM fine-tuned in a shared grapheme pivot space with only rendered synthetic exposure to the target, and that a stock-BPE ablation collapses this transfer, isolating output-space structure

as the mechanism. The campaign ran under a frozen preregistration whose primary hypothesis we falsified and report; the surviving two-factor account — synthetic budget plus pivot coverage — forecasts transfer to $\sim \pm 4.18$ WRR from quantities available before training, with its record of committed-in-advance hits and misses auditable by commit hash. Beyond the specific results, we offer the recipe — structural pivot, verified rendering, prospective evaluation — as a template: for the long tail of the world’s scripts, predictable zero-real-image transfer may matter more than another point of accuracy on scripts that are already read.

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